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Software and Hardware Tools

1. Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

Answer:

Steps:

1. Download and install **Wireshark** from its official website.
2. Open Wireshark and select your active network (WiFi/Ethernet).
3. Click **Start Capture**.
4. Open **Instagram or YouTube** and browse for 2 minutes.
5. Go back to Wireshark and click **Stop Capture**.
6. Save the file as **.pcap** (example: `network_capture.pcap`).

Top 3 Common Protocols You May See:

- * **HTTPS** – Used for secure browsing (Instagram, YouTube data transfer).
- * **DNS (Domain Name System)** – Converts website names into IP addresses.

** **TCP (Transmission Control Protocol)** – Ensures data is delivered correctly and in order.*

Explanation:

Wireshark shows all network traffic happening on your device. When you browse websites, multiple protocols work together to load videos, images, and data securely.

2. Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

Answer:

Steps:

1. Open Command Prompt.

2. Type:

``cmd

ping www.spotify.com

ping www.zomato.com

``

*3. Note the ****average response time (ms)**** from both results.*

Example Observation:

** Spotify average response time: **25 ms***

** Zomato average response time: **40 ms***

Conclusion:

*In this example, **Spotify responds faster** because it has lower latency (less response time).*

3. Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.

Answer:

Steps (Windows):

*1. Press ****Ctrl + Shift + Esc**** to open Task Manager.*

*2. Go to the ****Processes**** tab.*

3. Look for:

** Google Chrome*

** WhatsApp Desktop*

** Any browser (Edge, Firefox, etc.)*

Example Output:

** **Google Chrome.exe** – 300 MB to 1200 MB
(depends on tabs)*

** **WhatsApp.exe** – 150 MB to 300 MB*

** **msedge.exe (Microsoft Edge)** – 200 MB to 800 MB*

Explanation:

Each application uses memory (RAM) depending on how many tabs or processes are running. More tabs = more memory usage.

4. Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network.

Answer:

Steps:

- 1. Install **Angry IP Scanner** or **Advanced IP Scanner**.*
- 2. Open the software.*
- 3. Enter your local network range (example:
192.168.1.1 – 192.168.1.255).*
- 4. Start scanning.*
- 5. Wait for results.*

Example Output (Sample):

** **192.168.1.1** – WiFi Router*

** **192.168.1.5** – My Laptop*

** **192.168.1.8** – Mobile Phone*

5. Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

Answer:

1. Cable Tester

Used to check whether a LAN cable is working properly or has broken wires inside.

2. Crimping Tool

Used to attach RJ45 connectors to Ethernet cables while making or repairing network cables.

3. Multimeter

Used to measure electrical voltage, current, and continuity in network hardware and power supplies.

Explanation:

These hardware tools help network engineers find physical problems like broken cables, power issues, or faulty connections in a network setup.